

KOMMURI PRATAP REDDY INSTITUTE OF TECHNOLOGY

(UGC-Autonomous Institution)
Ghanpur(V), Ghatkesar(M), Medchal(D), Hyderabad-500088

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING (AI&ML)

MAJOR PROJECT DOCUMENTATION

HOMOMORPHIC-ENCRYPTION-BACKED PRIVACY-PRESERVING GenAI PIPELINE

*Encrypted machine-learning inference for privacy-sensitive healthcare and financial data – built
with TenSEAL / Microsoft SEAL and scikit-learn*

Under the Guidance of
Mrs. I. Vasantha Kumari
Department of CSE (AI&ML)

Presented by

Name	Roll Number
V. Manoj	23RA1A6654
V. Nandu	23RA1A6658
M. Nagesh Goud	23RA1A6657
U. Manoj	23RA1A6653

1. UML Diagrams

UML (Unified Modeling Language) diagrams describe the structure and behaviour of the system visually. Four diagrams are used here: the use case diagram (what the system does and for whom), the class diagram (static structure), the sequence diagram (order of messages over time) and the activity diagram (overall workflow).

1.1 Use Case Diagram

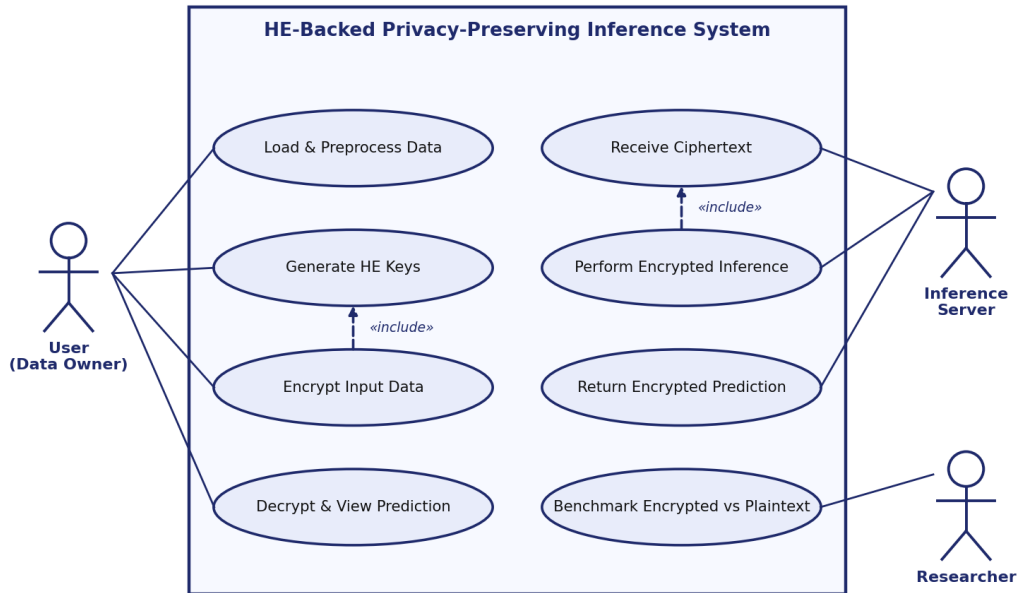


Figure 1: Use case diagram of the HE-backed inference system

The use case diagram shows the functional requirements from the point of view of the people and systems that interact with the pipeline. The rectangle is the system boundary, the ellipses are use cases, and the dashed arrows marked «include» show a use case that always relies on another one. Three actors take part: the User (data owner), who holds the sensitive data and the secret key; the Inference Server, which hosts the model but only ever handles ciphertext; and the Researcher, who evaluates what HE costs.

Use Case	Actor	Description
Load & Preprocess Data	User	Clean, scale and split the sensitive dataset.
Generate HE Keys	User	Create the public context and the secret key.
Encrypt Input Data	User	Turn features into ciphertext (includes key generation).
Decrypt & View Prediction	User	Decrypt the returned prediction and read the result.
Receive Ciphertext	Server	Accept ciphertext and public context only.
Perform Encrypted Inference	Server	Run the model on ciphertext (includes receiving it).
Return Encrypted Prediction	Server	Send the encrypted result back to the client.
Benchmark Encrypted vs Plaintext	Researcher	Compare accuracy, latency and overhead.

1.2 Class Diagram

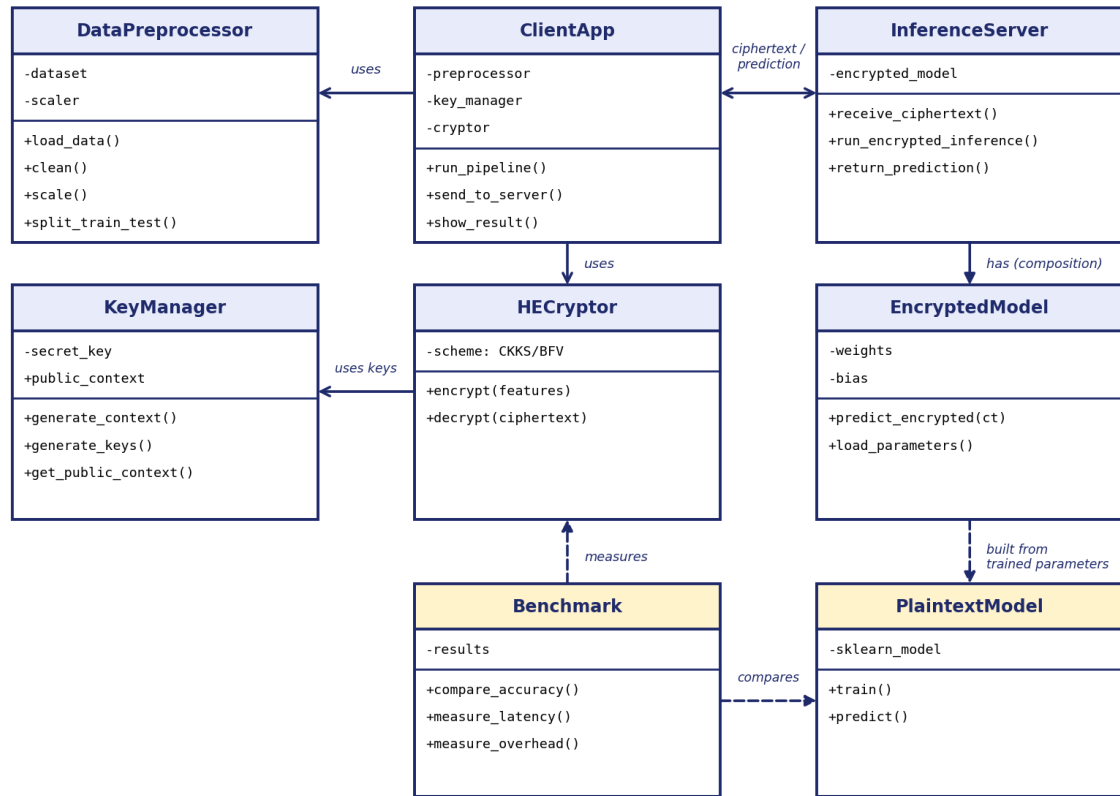


Figure 2: Class diagram of the pipeline

The class diagram shows the static structure of the system: eight classes with their attributes (top compartment) and operations (bottom compartment). Solid arrows with the word "uses" are associations, the double-headed arrow is the two-way exchange of ciphertext and encrypted predictions, the dashed arrows are dependencies, and "has (composition)" means the InferenceServer owns its EncryptedModel. The yellow classes belong to the evaluation side.

Class	Responsibility
DataPreprocessor	Loads, cleans, scales and splits the dataset on the client.
KeyManager	Generates the HE context and key pair; keeps the secret key on the client.
HECryptor	Encrypts features into ciphertext and decrypts the final prediction (CKKS / BFV).
ClientApp	Orchestrates the client side: preprocess, encrypt, send, decrypt, show result.
InferenceServer	Receives ciphertext, runs encrypted inference and returns the encrypted prediction.
EncryptedModel	Holds weights and bias and evaluates the model on ciphertext.
PlaintextModel	The scikit-learn model that is trained normally and used as the baseline.
Benchmark	Compares encrypted and plaintext results: accuracy, latency and overhead.

1.3 Sequence Diagram

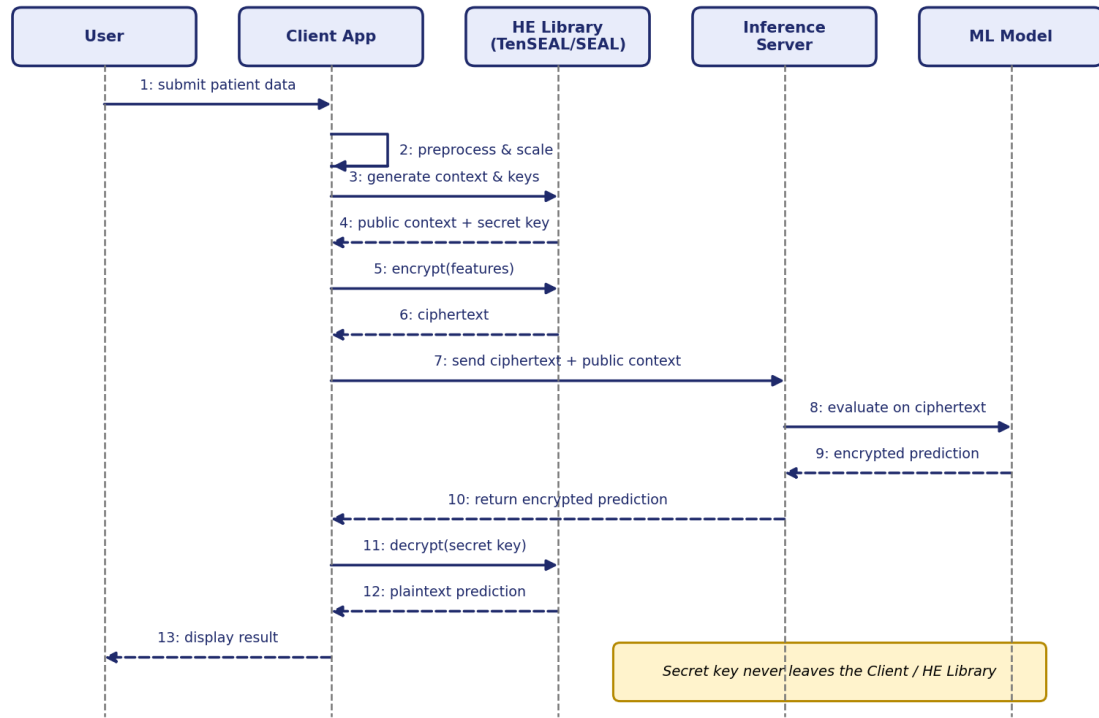


Figure 3: Sequence diagram of one encrypted prediction request

The sequence diagram shows the order of messages between the five participants for a single prediction. Time runs downward; solid arrows are calls and dashed arrows are returns. The messages fall into four phases:

- Key setup (1-4): the user submits data, the Client App preprocesses it and asks the HE library for a context and keys; the public context and secret key come back to the client.
- Encryption (5-6): the features are encrypted by the HE library and returned as ciphertext.
- Inference (7-10): only the ciphertext and public context are sent to the server, which evaluates the ML model on ciphertext and returns an encrypted prediction.
- Decryption (11-13): the client decrypts with the secret key and displays the plaintext result to the user.

The key point is that the server and model appear only in messages 7-10, where nothing but ciphertext is exchanged.

1.4 Activity Diagram

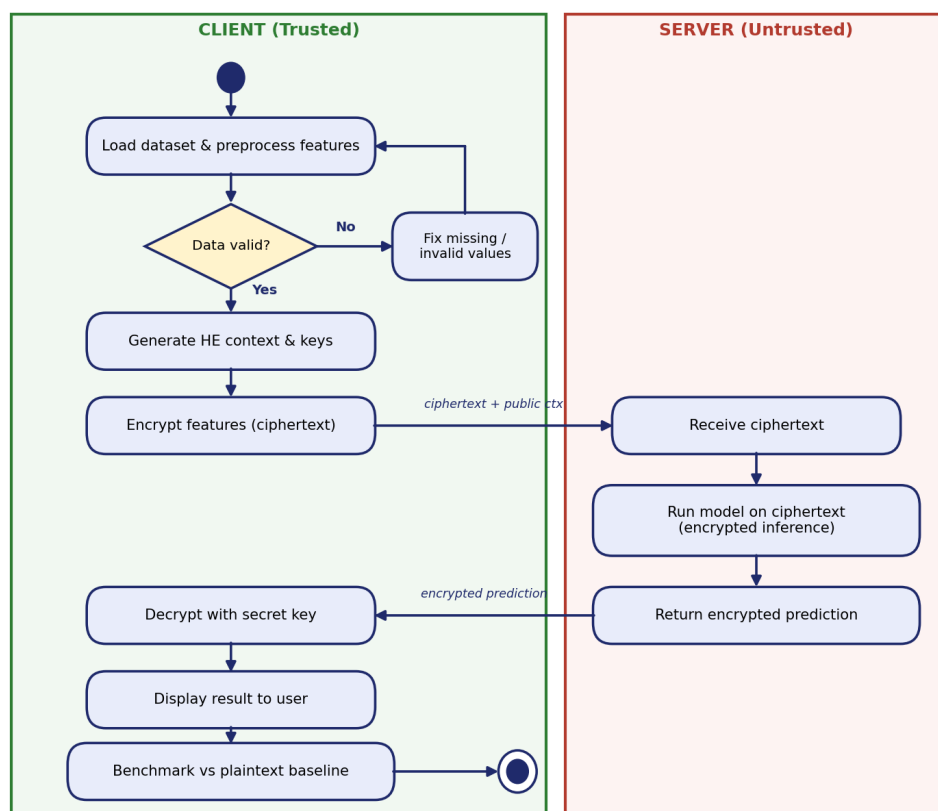


Figure 4: Activity diagram of the end-to-end workflow

The activity diagram shows the workflow from start to finish, split into two swimlanes that mark the trust boundary: the client (trusted) and the server (untrusted).

- The flow starts on the client: the dataset is loaded and preprocessed, and a decision node checks that the data is valid. If not, missing or invalid values are fixed and the data is checked again.
- Once the data is valid, the client generates the HE context and keys and encrypts the features.
- Control then passes to the server, which receives the ciphertext, runs the model on it and returns an encrypted prediction. The server lane contains no decryption step.
- Control returns to the client, which decrypts the prediction with the secret key and shows it to the user.
- Finally the encrypted result is benchmarked against the plaintext baseline, and the flow ends.